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Driver Digital Twin for Online Recognition of Distracted Driving Behaviors

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***Abstract*—Deep learning has been widely utilized in intelligent vehicle systems, particularly in the field of driver distraction detection. However, existing methods in this application tend to focus solely on appearance or cognitive state as indicators of distraction, while neglecting the significance of temporal modeling in accurately identifying driver actions. This oversight can result in limitations such as difficulty in comprehending context, incapability to recognize gradual changes, and failure to capture complex behaviors. To address these limitations, this paper introduces a new framework based on the concept of Driver Digital Twin (DDT). The DDT framework serves as a digital replica of the driver, capturing their naturalistic driving data and behavioral models. It consists of a transformer-based driver action recognition module and a novel temporal localization module to detect distracted behaviors. Additionally, we propose a pseudo-labeled multi-task learning algorithm that includes driver emotion recognition as supplementary information for recognizing distractions. We have validated the effectiveness of our approach using three publicly available driver distraction detection benchmarks: SFDDD, AUCDD, and SynDD2. The results demonstrate that our framework achieves state-of-the-art performance in both driver action recognition and temporal localization tasks. It outperforms the leading methods by 6.5 and 0.9 percentage points on SFDDD and AUCDD, respectively. Furthermore, it ranks in the top 5% on the SynDD2 leaderboard.**

***Index Terms*—Digital twin, driver behavior modeling, automated vehicle, driver distraction detection, temporal action localization.**

I. Introduction

The continuous advancements in intelligent vehicles are revolutionizing our transportation systems, offering exciting opportunities for improved safety, mobility, and environmental sustainability. As automated vehicles evolve, it becomes increasingly important for them to demonstrate enhanced intelligence, safety features, and personalization options. This rapidly changing landscape highlights the need to redefine the interaction between humans and machines in shared driving scenarios, which poses a significant challenge.

To address this challenge, the concept of a Digital Twin [1] introduces the idea of a Driver Digital Twin (DDT) as a potential solution [2], [3]. The DDT draws inspiration from the parallel intelligence approach in the cyber-physical-social space, considering interactions among vehicles, human drivers, and information [4]–[6]. Building upon our previous research on Mobility Digital Twin (MDT) [7] and Metamobility [8], the DDT serves as a digital replica of the driver, encapsulating their naturalistic driving data and behavior models [9]. Utilizing this digital representation, a DDT framework can offer various micro-services, such as driver distraction or drowsiness detection and driver intent prediction.

Among various challenges to road safety, driver distraction stands as a significant concern. Defined as any activity that diverts attention from the task of driving, distractions range from cell phone usage to eating, drinking, and conversing with passengers. According to the National Highway Traffic Safety Administration (NHTSA), distraction-induced accidents accounted for 8.1% of the 38,824 motor vehicle-related fatalities in the U.S. in 2020 [10]. This problem becomes even more prominent with the growing reliance on automated driving systems, especially those at SAE Level 3 [11]. These systems allow drivers to disengage from direct vehicle control but necessitate their readiness to resume control when needed. Given the propensity for drivers to become less attentive and more susceptible to distractions when not actively controlling the vehicle, addressing driver distraction becomes a critical function of the DDT framework.

Driver action recognition is a widely adopted approach used to identify and mitigate driver distraction. This methodology involves continuously monitoring drivers using non-intrusive sensors such as RGB cameras and infrared sensors, combined with advanced deep learning techniques to recognize and categorize driver actions. Recent studies have made significant progress in recognizing driver actions [12]–[14]. However, it is important to note that most of these efforts primarily focus on classifying distracted driving behaviors using single-frame camera inputs.

The emphasis on single-frame analysis can have drawbacks as it overlooks the crucial aspect of temporal action modeling, which is essential when dealing with video or camera stream data. The absence of temporal action modeling can introduce limitations and challenges to driver action recognition systems. Without considering the temporal dimension of driver behaviors, these systems may struggle to understand the context in which actions occur. Additionally, they may be less effective in recognizing gradual transitions between actions or identifying complex behaviors that evolve over time [15], [16].

Another limitation of current methodologies is the limited consideration given to the driver's cognitive state, which is a crucial factor influencing distraction. While physiological

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sensors such as electroencephalographic (EEG) signals could provide insights into the driver's cognitive state [17], [18], it is crucial to carefully evaluate the impact of these sensors on natural driving behavior, as their use may introduce certain considerations regarding user comfort and potential disruptions. For instance, some studies have highlighted the need to account for the potential influence of these sensors on distraction detection results [2]. Additionally, existing driver distraction datasets lack the necessary mental state labels required for training models to learn to recognize mental states from appearance sensor data, necessitating the adoption of self-learning methods to address this underexplored challenge.

To address these gaps, this paper first explores the important role of the DDT framework in identifying driver distractions. It proposes a unified transformer-based approach for recognizing driver actions and accurately determining their start and end times. This framework allows for the recognition of various driver actions and enhances the precision of identifying distracted behaviors.

Next, we explore the potential of using emotional states as an additional indicator of distraction. This approach eliminates the need for physiological sensors and additional labeling efforts. We introduce a pseudo-labeled multi-task learning (MTL) algorithm that leverages the inductive information from driver emotion recognition to improve the identification of distracted behaviors.

The proposed framework's performance in the recognition task is evaluated on two widely-used datasets: State Farm Distracted Driver Detection (SFDDD) [19] and the American University in Cairo Distracted Driver Dataset (AUCDD) [20], [21]. SFDDD consists of 22K labeled images from 26 drivers, while AUCDD contains 10K training images and 1K testing images from 44 drivers. Both datasets cover the same 10 real-world driving actions. For the temporal localization task, our framework is evaluated using the SynDD2 dataset [22], which includes 150 training video clips and 30 testing videos, recorded at a resolution of 1920×1080 and synchronized across three camera views. The proposed framework achieves state-of-the-art performance on all three benchmarks, surpassing the leading methods by 6.5 and 0.9 percentage points on SFDDD and AUCDD, respectively. It also ranks 5th on the public leaderboard of the SynDD2 test set.

The paper's primary contributions can be summarized as follows:

- A novel transformer-based framework designed to accurately recognize and temporally localize driver actions.
- A new temporal localization algorithm called "Election" to refine the results obtained from the transformer network. This highlights the importance of temporal modeling in distraction detection.
- A multi-task learning algorithm that incorporates pseudo-labels and includes driver emotion recognition as a concurrent task. This algorithm improves the framework's reliability by incorporating emotion states as supplementary information.
- Validation of the proposed framework's effectiveness by achieving state-of-the-art performance across three public driver distraction detection benchmarks.

This paper is structured as follows: In Sec. II, we provide a concise review of relevant literature. In Sec. III, we elaborate on our proposed transformer-based method for driver action recognition and temporal localization. In Sec. IV, we present a detailed exposition of the experiments conducted for both tasks, along with comprehensive analyses of their results. Sec. V furnishes qualitative results, elucidating the interpretability of our proposed framework. Finally, in Sec. VI, we conclude the paper by summarizing the significant findings and discussing potential directions for future research.

II. Related Work

A. Digital Twins and Parallel Intelligence

Related literature in the fields of "parallel driving" or "parallel intelligence" has direct implications for the DDT concept explored in our study. In a seminal work by Wang [23], the concept of parallel transportation was introduced, defining parallel control and management of transportation as "a data-driven approach for modeling, analysis, and decision making that takes into account both the engineering and social complexity in its processes".

Since then, numerous subsequent studies have been conducted in this research domain [4], [24]–[32]. These studies have further expanded the understanding of parallel driving and its potential applications. For instance, Wang [6] proposed a comprehensive parallel driving framework that considers the cloud-based cyber-physical-social system. This framework models the physical world, mental world, and artificial world as three parallel levels, taking into account the interactions among connected vehicles, human drivers, and information. The exploration of parallel driving and parallel intelligence has opened up new avenues for research and development in transportation systems [5], [33], [34]. By incorporating these principles, the efficiency, safety, and sustainability of transportation networks can be enhanced. Furthermore, the integration of connected vehicles, human drivers, and information in a parallel framework allows for a more comprehensive understanding of transportation dynamics and facilitates better decision-making processes.

B. Driver Distraction Detection

Driver Distraction Detection constitutes one of the most important components of a DDT framework. To detect driver distractions, both intrusive and non-intrusive sensors are employed, in conjunction with machine learning (ML) algorithms to identify driver distractions. A common approach involves the recognition of distractions based on drivers' actions [12], [13]. These studies typically utilize vision-based sensors and employ deep convolutional neural networks (CNNs) to recognize driver actions. Another method focuses on drivers' head movements and gaze patterns as indicators of distractions [35], [36]. Similarly, these studies treat distraction detection as a classification task, leveraging CNN-based models that use driver facial image features as input rather than full-body images. However, these methods bear certain limitations. Firstly, CNN-based models primarily use convolution layers, assuming local feature interactions. This design limits their ability to model global interactions, such as the interaction between a driver and a passenger, which can play a significant role in recognizing driver distractions. Secondly, these methods mainly concentrate on the action classification on single images in the spatial dimension and neglect the temporal dimension. This oversight could undermine their effectiveness when applied to onboard cameras if the temporal information dependency is not utilized. Finally, these methods overlook the driver's mental state, a significant indicator of distraction. Although some studies have used EEG signals to measure brain activity for the detection of driver distractions [17], [18], the use of intrusive physiological sensors can lead to discomfort, potentially distorting natural driving behavior and subsequently biasing distraction detection results [2].

In this paper, we propose a unified framework to address these limitations. Our framework introduces a new temporal analysis technique, which goes beyond static frame analysis and enables the processing of continuous driver behavior over time. Additionally, we integrate cognitive analysis into our framework to consider the significant impact of a driver's mental state on distraction, without the need for intrusive sensors. Furthermore, we incorporate attention mechanisms into our framework, leveraging their effectiveness in filtering and highlighting critical information. This mechanism prioritizes relevant features and sequences within the data, enabling a more focused and effective analysis of potential distractions.

C. Driver Emotional State Monitoring

Driver Emotional State Monitoring is another fundamental function of a DDT framework, given that a driver's emotional state significantly influences driving performance [37]. Over the past decades, emotion recognition has garnered substantial attention, with related research being broadly divided into two categories.

One approach employs wearable physiological sensors to detect a human driver's emotional state [38]. The second is a vision-based method [39], which is founded on the fact that emotions can be inferred from facial expressions [2], and is also referred to as Facial Expression Recognition (FER). In this paper, FER is applied to drivers' facial images to assess their emotional state. This offers additional insight for detecting driver distractions through the application of MTL.

D. Multi-Task Learning and Self-Training

MTL is an inductive transfer technique that aims primarily at enhancing the generalization performance of learning models. Traditional machine learning paradigms usually focus on learning one task at a time. However, Caruana [40] argued that such an approach might be suboptimal as it overlooks a potentially rich source of information embedded in many real-world scenarios, i.e., the insights contained within the training signals of other related tasks. If these tasks can leverage shared learnings, it might be more beneficial to concurrently learn multiple tasks. In the field of computer vision, a prevalent approach to MTL is to utilize a single encoder that learns a common representation, which is then processed by several task-specific decoders [41]–[45]. In this paper, we adopt a similar strategy, training a primary backbone model in conjunction with several small task-specific heads. Self-training is a method used to incorporate unlabeled data into a supervised learning task [46], which represents one of the earliest semi-supervised learning techniques, generating pseudo labels for unlabeled data with the aid of a supervised model.

III. METHODOLOGY

In this section, we present the design of our proposed transformer-based approach for the recognition and temporal localization of driver actions.

A. Problem Definition

a) Action Recognition: In the given context, the recognition process is targeted at accurately categorizing the driver's actions. The input data for this task either comes in the form of a single-frame image or a video snippet of a predetermined length, which we refer to as a super-frame. These visuals are collected from one or more in-cabin camera perspectives. The intended result is the precise identification of the specific action undertaken by the driver.

b) Temporal Localization: Temporal localization builds upon the principle of recognition and is designed to function on a live camera stream or an untrimmed video. The goals of this process are twofold: first, to accurately identify the various actions enacted by the driver; second, to determine the exact start and end times of each action. This thorough temporal analysis assists in distraction detection, providing an in-depth understanding of their duration and order within the driving environment.

We introduce several symbols in this section that will be utilized throughout the rest of this paper. Specifically, we define the number of camera views as M. This leads us to express a multi-view untrimmed video as $\mathbf{v} = (\mathbf{v}_1, ..., \mathbf{v}_t, ..., \mathbf{v}_T)$, where $\mathbf{v}_t$ is a multi-view frame. Specifically, $\mathbf{v}_t$ is defined as:

$$\mathbf{v}_t = \left\{\mathbf{v}_{t,m} \in \mathbb{R}^{C\times H_m \times W_m}\right\}_{m=1}^{M}, \tag{1}$$

where H_m and W_m refer to the height and width of the input image obtained from view m. C stands for the number of color channels. Moreover, we represent a multi-view super-frame as

$$\mathbf{x} = (\mathbf{v}_1, ..., \mathbf{v}_t, ..., \mathbf{v}_{T_{\text{sf}}})_{t=1}^{T_{\text{sf}}}, \tag{2}$$

where T_{sf} is the number of frames in a super-frame.

Let $\mathcal{C}$ be the set of predefined action categories. We use $\mathbf{y}$ to denote the ground-truth set of driver actions that occurred in the video. Each element i within the ground-truth set is expressed as $\mathbf{y}_i = (s_i, e_i, c_i)$, where s_i indicates the start time, e_i signifies the end time and $c_i \in \mathcal{C}$ corresponds to the associated action label.

B. Recognizing Driver Actions

Accurate recognition of distracted driving behaviors requires a robust model capable of understanding video content effectively. To achieve this, we employ a ViT as the backbone of our recognition module, considering its demonstrated

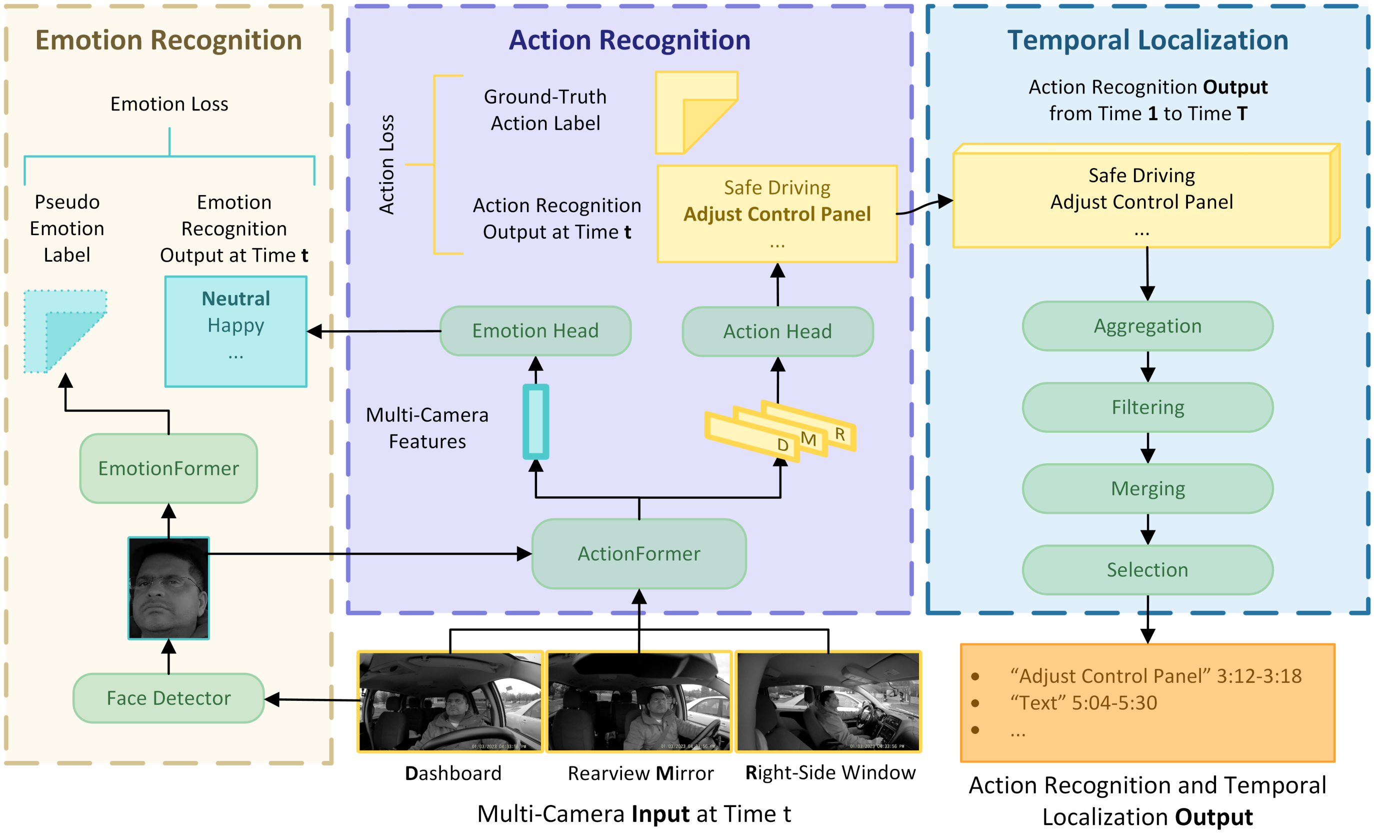

Fig. 1. Architecture of the proposed unified framework for driver action recognition and temporal localization. The model consists of three main components: 1) Emotion-Recognition augmented Multi-task learning module with pseudo labeling **(Left)**, 2) Transformer-based action recognition module **(Center)**, and 3) Temporal Localization module, which includes aggregation, filtering, merging, and selection processes **(Right)**.

proficiency in processing long-range dependencies and its exceptional performances across various action recognition benchmarks. In our framework, we refer to this model as the ActionFormer. The entire recognition module is visually represented in the middle part of Fig. 1.

The ActionFormer takes a super-frame $\mathbf{x}$ as input. In cases where $\mathbf{x}$ contains multiple views, each view is individually fed into the ActionFormer, with shared weights to reduce complexity. This process produces a set of extracted features from the camera inputs. Subsequently, an action head is applied to these features, generating a probability distribution over the action categories $\mathcal{C}$.

To train the ActionFormer, we randomly select a super-frame from each action segment in the training set, where each segment covers the full duration of a single action. These selected super-frames collectively form a training epoch, ensuring a diverse range of training examples. Notably, during training, the sampling process is conducted independently for each camera view. During inference, however, the model processes the multi-view input and simultaneously considers all available camera views to generate predictions. This approach allows the model to learn from individual views during training and leverage the collective information from multiple views during inference, resulting in robust recognition of driver actions.

C. Temporally Localizing Driver Actions

To facilitate temporal action localization, we adopt a sliding window technique to process an untrimmed multi-view video or a live multi-view camera stream. Specifically, we employ a window size identical to ActionFormer's input size and glide this window across the entire length of the video. For every window, we input the corresponding frames from all three camera views into the ActionFormer, generating a probability matrix $\mathbf{p} \in \mathbb{R}^{T\times|\mathcal{C}|\times M}$, where T and $|\mathcal{C}|$ represent the video's duration and the total number action categories, respectively.

To refine the preliminary results obtained from the action recognition module, we propose a novel algorithm, referred to as Election. This algorithm takes the probability matrix $\mathbf{p}$ as its input. The proposed method includes four steps:

a) Aggregation (AGG).: The first step applies a convolution operation to the input probability matrix to gather information from various camera perspectives. Specifically, the operation is defined as $\mathbf{p}' = \mathbf{p} \times \omega$, where $\mathbf{p}' \in \mathbb{R}^{T\times|\mathcal{C}|}$ represents the aggregated probability scores, $\mathbf{p}$ is the original probability matrix, and $\omega \in \mathbb{R}^{|\mathcal{C}|\times M}$ denotes the convolution kernels. In this equation, each entry of the new matrix $\mathbf{p}'$, denoted as $\mathbf{p}'_{t,c}$, is computed as the dot product of the corresponding row in $\mathbf{p}$ and the column in ω, symbolically represented as $\mathbf{p}'_{t,c} = \sum_{m=1}^{M} \omega_{c,m} \cdot \mathbf{p}_{t,c,m}$. This operation assigns different weights to each camera view m for each action category c, allowing for the integration of complementary information from different perspectives.

b) Filtering (FLTR).: In the second step, initial action candidates are designated by extracting continuous frames with probability scores exceeding a predefined threshold for each action category. Frames with probabilities below this threshold are discarded, thereby diminishing the likelihood of false positives.

c) Merging (MRG).: The third step merges temporally adjacent clips that have a minor gap between them (e.g., less than 0.5 seconds). This process aims to accommodate brief pauses that may occur during certain driver actions, ensuring a comprehensive depiction of each action.

d) Selection (SEL).: After the merging process, the average score of all merged candidates for each action category is computed. In situations where multiple candidates are present for an action category, the candidate with the highest average score is chosen as the final action candidate. The Election algorithm then outputs $|\mathcal{C}|$ final action candidates, one for each action category. This exhaustive evaluation ensures a more precise and reliable temporal localization of the driver's actions.

D. Enhancing Recognition via Multitask Learning with Emotion Recognition

Multitask Learning (MTL) functions as an inductive transfer technique that improves generalization performance by leveraging shared information across multiple related tasks [40]. In our approach, we integrate driver emotion recognition as a concurrent task to augment the efficacy of driver action recognition. This integration is grounded on the significant correlation between the driver's emotional state and their actions. For example, safe driving typically corresponds with a neutral emotional state, whereas interaction with a passenger may provoke various emotions.

Our proposed multitask self-learning algorithm is illustrated on the left side of Fig. 1. In the initial phase, we train a ViT model on a comprehensive FER dataset called AffectNet-7 [47] using supervised learning. This trained model, referred to as the EmotionFormer, is capable of accurately identifying drivers' facial expressions. Upon extracting face images from the input frames using a face detector, the EmotionFormer is utilized to assign pseudo-emotion labels to these driver face images. These pseudo labels are generated based on the model's predictions of the emotions conveyed by the detected faces. Subsequently, an enriched driver dataset is compiled, including both the actual driver action labels and the inferred emotion labels. Notably, each image from the primary dataset (see details in Sec. IV-A) is annotated with a pseudo-emotion label that falls into one of Ekman's six fundamental emotional categories [48], as well as an additional category for "neutral" emotions. This augmented dataset is subsequently leveraged to train the ActionFormer via MTL.

During MTL, distinct patch projection layers are applied to the face image $\mathbf{v}_{\text{face}} \in \mathbf{v}$ and the driver image $\mathbf{v}_{\text{driver}} \in \mathbf{v}$. The images $\mathbf{v}_m \in \mathbf{v}$ are divided into a grid of N_m patches, each with a size of $P^2 \cdot C$, where (P, P) is the patch size. These patches are then flattened and projected to D-dimensional tokens:

$$\mathbf{v}_m^{\text{vit}} = [\mathbf{v}_m^1 \mathbf{E}_m, ..., \mathbf{v}_m^{N_m} \mathbf{E}_m] + \mathbf{E}_m^{\text{pos}}, \quad m = 1, 2, ..., M, \quad (3)$$

where $\mathbf{E}_m \in \mathbb{R}^{(P^2 \cdot C) \times D}$ is the projection matrix, and $\mathbf{E}_m^{\text{pos}} \in \mathbb{R}^{N_m \times D}$ represents the position embedding. The embeddings from the driver and face images are then concatenated into a combined sequence $[\mathbf{v}_1^{\text{vit}}; \mathbf{v}_2^{\text{vit}}; \cdots; \mathbf{v}_M^{\text{vit}}]$, which is subsequently fed into the ActionFormer.

The ActionFormer then learns the fused driver behavior and emotion representation by stacking L Transformer blocks . A Multilayer Perceptron module and a Multihead Self-attention module are included in each block. Self-attention (SA) is the key component of Transformer blocks, in which, the input vector $\bar{x}$ is first transformed into three separate vectors: the query vector $\bar{q} = \bar{x}$, the key vector $\bar{k} = \bar{x}$, and the value vector $\bar{v} = \bar{x}$, all of the same dimension $\bar{q}, \bar{k}, \bar{v} \in \mathbb{R}^D$. After that, the attention scores are constructed by the following function:

$$\text{SA}(\bar{x}) = \text{softmax}\left(\frac{W_Q \bar{q} \cdot (W_K \bar{k})^\top}{\sqrt{D}}\right)(W_V \bar{v}) \quad (4)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D \times D}$ are learnable parameters of three linear projections.

During training, we optimize the ActionFormer by minimizing the joint loss function, $\mathcal{L}_{\text{joint}}$, which consists of two components: the action recognition loss, $\mathcal{L}_{\text{action}}$, and the emotion recognition loss, $\mathcal{L}_{\text{emotion}}$. The action recognition loss measures the disparity between the predicted action labels ($\hat{y}^c$) and the ground-truth action labels (y^c), while the emotion recognition loss quantifies the difference between the predicted emotion labels ($\hat{y}^e$) and the pseudo-emotion labels ($\tilde{y}^e$). These losses are calculated using the standard cross-entropy loss function. Therefore, the joint loss function is defined as:

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{emotion}} \quad (5)$$

$$\mathcal{L}_{\text{action}} = -\sum_{i=1}^{N} y_i^c \log(\hat{y}_i^c) \quad (6)$$

$$\mathcal{L}_{\text{emotion}} = -\sum_{i=1}^{N} \tilde{y}_i^e \log(\hat{y}_i^e), \quad (7)$$

where N represents the number of samples in the dataset. By optimizing the joint loss function, our ActionFormer learns to simultaneously recognize driver actions and emotions, leveraging the shared information between the two tasks during training. This integration of emotion recognition enhances the effectiveness of driver action recognition.

IV. Experiments

A. Benchmark Datasets

Our proposed framework's performance in the recognition task is evaluated on two widely-used, publicly accessible datasets: State Farm Distracted Driver Detection (SFDDD) and the American University in Cairo Distracted Driver Dataset (AUCDD). SFDDD comprises 22,424 labeled images showcasing 26 drivers, as captured by a dashboard camera with a resolution of 640×480 [19]. On the other hand, AUCDD

TABLE I
LIST OF DRIVER ACTIONS IN THE BENCHMARKS: L, R, D, AND P REPRESENT LEFT, RIGHT, DRIVER, AND PASSENGER, RESPECTIVELY.

(a) SFDDD and AUCDD

ID	Description	ID	Description
0	Safe Driving	5	Adjust control panel
1	Phone Call (R)	6	Drinking
2	Phone Call (L)	7	Hair or makeup
3	Text (R)	8	Reaching behind
4	Test (L)	9	Talk to pax

(b) SynDD2

ID	Description	ID	Description
0	Forward Driving	8	Adjust control panel
1	Drinking	9	Pick up from floor (D)
2	Phone Call (R)	10	Pick up from floor (P)
3	Phone Call (L)	11	Talk to pax at the right
4	Eating	12	Talk to pax at backseat
5	Text (R)	13	Yawning
6	Text (L)	14	Hand on head
7	Reaching behind	15	Singing or dancing

contains 10,555 training images and 1,123 testing images with a higher resolution of 1920×1080, involving data from 44 drivers, split into 38 for training and 6 for testing [20], [21]. Both datasets include the same ten real-world driving actions, as detailed in Tab. I(a).

Two common train/test split strategies are employed for the SFDDD dataset. The first approach follows an image-based split for training and testing [49]–[52]. However, this method can create an unwanted correlation between training and testing data, as successive video frames might be divided between the two sets, inadvertently simplifying the recognition task. As a more robust alternative, the dataset can be partitioned by the driver, thereby ensuring images from a single driver do not straddle both the training and test data [53]. We report our results employing both partition methods for the SFDDD dataset. For the image-based split, we use a $70\%/30\%$ division for training and testing, respectively. For the driver-based split, we randomly select 18 out of 28 drivers for training, with the remaining set aside for testing. We maintain the original split-by-driver setup for the AUCDD dataset.

For the temporal localization task, we assess our framework using the SynDD2 dataset [22]. The SynDD2 dataset is comprised of 150 training video clips and 30 testing videos. Captured at a resolution of 1920×1080, these videos were recorded at a rate of 30 frames per second, with manual synchronization across three camera views [54]. These cameras were placed near the dashboard, rearview mirror, or right-side window. Each video, approximately 9 minutes long, includes 16 different distracted driving activities as detailed in Tab. I(b). These activities were performed by the driver in a random sequence for varying durations, both with and without appearance blocks such as hats or sunglasses. Each driver is featured in six videos: three with an appearance block and three without. This dataset also forms part of the 7th AI City Challenge Track 3 [55].

B. Evaluation Metrics

In evaluating the recognition task, we utilize two standard evaluation metrics: average accuracy and negative log-likelihood (NLL). These metrics allow for consistent comparisons with previous works.

The average accuracy is calculated by considering the ratio of correctly classified super-frames to the total number of super-frames. To ensure equal treatment of all categories, we adopt a balanced approach and calculate the per-class accuracy. Let N_c be the number of super-frames in category c, and N be the total number of samples, where $N = \sum_{c \in \mathcal{C}} N_c$. The average accuracy is computed as:

$$\text{Accuracy} = \frac{1}{N}\left(\sum_{c \in \mathcal{C}} N_c \times \text{accuracy}_c\right), \tag{8}$$

where accuracy_c represents the accuracy of category c.

The NLL is employed to assess the quality of the predicted probabilities. It is computed as the average logarithmic loss across all samples:

$$\text{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log(\Pr(y_i|\mathbf{x}_i)), \tag{9}$$

where $\mathbf{x}_i$ represents the input super-frame, y_i is the ground truth label, and $\Pr(y_i|\mathbf{x}_i)$ denotes the predicted probability of the correct label.

For the temporal localization task, we apply the activity overlap score utilized in the 7th AI City Challenge [55] as the evaluation metric. Let $\hat{\mathbf{y}}$ represent the set of N_p predictions made by the framework, where N_p is the cardinality of $\mathcal{C}$. To compute the overlap score, a bipartite matching is established between the ground-truth set $\mathbf{y}$ and the predicted set $\hat{\mathbf{y}}$. This matching yields a permutation of N_p elements, $\sigma \in \mathfrak{S}_{N_p}$, which produces the highest overlap score:

$$\hat{\sigma} = \text{argmax}_{\sigma \in \mathfrak{S}_{N_p}} \sum_{i=1}^{N_p} os(\mathbf{y}_i, \hat{\mathbf{y}}_{\sigma(i)}), \tag{10}$$

where $os(\mathbf{y}_i, \hat{\mathbf{y}}_{\sigma(i)})$ represents the pairwise overlap score between the ground truth $\mathbf{y}_i$ and a prediction indexed as $\sigma(i)$. A match is considered valid if $\mathbf{y}_i$ and $\hat{\mathbf{y}}_{\sigma(i)}$ belong to the same class ($c_i = \hat{c}_{\sigma(i)}$), and the start and end times $\hat{s}_{\sigma(i)}$ and $\hat{e}_{\sigma(i)}$, respectively, fall within a 10-second range before or after the ground-truth activity's start (s_i) and end (e_i) times. The overlap score between each matched pair of activities is computed as the proportion of their time intersection to their time union, defined as:

$$os(\mathbf{y}_i, \hat{\mathbf{y}}_{\sigma(i)}) = \frac{\max(\min(e_i, \hat{e}_{\sigma(i)}) - \max(s_i, \hat{s}_{\sigma(i)}), 0)}{\max(e_i, \hat{e}_{\sigma(i)}) - \min(s_i, \hat{s}_{\sigma(i)})}. \tag{11}$$

If a ground-truth activity lacks a match, or a predicted activity does not have a corresponding ground-truth match, an overlap score of 0 is assigned. The final score is then calculated by averaging the overlap score across all matched and unmatched activities.

TABLE II

COMPARISON BETWEEN THE PROPOSED FRAMEWORK WITH SEVERAL SOTA METHODS ON THE RECOGNITION TASK. THE BEST METHOD AMONG EACH SETTING IS HIGHLIGHTED IN **BOLD**. ↓ INDICATES LOWER IS BETTER. ↑ INDICATES HIGHER IS BETTER. * RESULTS FROM THE ORIGINAL PAPERS.

Experiment	Method	Accuracy (↑)	NLL (↓)
AUCDD	GA-Weighted Ensemble* [20]	0.9006	0.6400
	ADNet* [56]	0.9022	–
	C-SLSTM* [57]	0.9270	0.2793
	Our Framework	**0.9359**	**0.2399**
SFDDD (Split-by-Driver)	DD-RCNN* [53]	0.8600	**0.3900**
	Our Framework	**0.9251**	0.3972
SFDDD (Split-by-Image)	DDR-ViT-finetuned* [52]	0.9750	–
	ViTConv* [49]	0.9790	0.0800
	Inception+ResNet+HRNN* [50]	0.9930	–
	LWANet* [51]	0.9937	0.0260
	Our Framework	**0.9963**	**0.0171**

TABLE III

ABLATION STUDY COMPARING THE EFFECTIVENESS OF THE PROPOSED MTL STRATEGY WITH EMOTION RECOGNITION.

Dataset	MTL	Accuracy (↑)	NLL (↓)
SFDDD	✗	0.9036	0.5355
	✓	**0.9251**	**0.3972**
AUCDD	✗	0.9092	0.2895
	✓	**0.9359**	**0.2399**

TABLE IV

ABLATION STUDY EXAMINING THE EFFECTIVENESS OF EACH STEP OF THE ELECTION ALGORITHM. THE SCORES ARE OBTAINED BY SUBMITTING THE INFERENCE RESULTS TO THE EVALUATION SERVER.

Step				Overlap Score
SEL	FLTR	MRG	AGG	
✓	✗	✗	✗	0.4683
✓	✓	✗	✗	0.5347
✓	✓	✓	✗	0.5565
✓	✓	✓	✓	**0.5921**

C. Implementation Details

For the SFDDD and AUCDD datasets, we employ the standard Vision Transformer (ViT-B/16) [58], pre-trained on the ImageNet [59] dataset, as our backbone recognition model. This model is configured to receive single-frame images as inputs, with in-cabin images resized to dimensions of 224×224 and face images adjusted to 32×32. For the SynDD2 dataset, we employ the MViTv2-B [60], pretrained on the Kinetics-700 [61] dataset as the backbone model. This model processes super-frames, each comprising $T_{\text{sf}} = 16$ frames sampled at a stride of 4 frames. All three camera views are configured to maintain a consistent input size of 448×448. We did not implement the MTL strategy for the SynDD2 dataset as we aim to provide a fair comparison with other methods in the 7th AI City Challenge where the use of external data is disallowed.

Training across all datasets was performed using the AdamW optimizer [62], supplemented by a cosine scheduler. The batch sizes for the SFDDD, AUCDD, and SynDD2 datasets were configured at 256, 256, and 2, respectively. Guided by the linear scaling rule [63], we empirically established the base learning rates for each dataset, setting them to 3×10^{-4}, 6×10^{-4}, and 5×10^{-6}, respectively.

D. Results on Recognition

In this section, we present performance comparisons for the recognition task between our proposed framework and state-of-the-art (SOTA) approaches. Since the existing literature primarily reports performance in terms of accuracy (Eq. (8)) and NLL (Eq. (9)), our comparison focuses on these two metrics. The comparisons, detailed in Tab. II, highlight the effectiveness of our framework in recognizing driver distractions.

Our framework demonstrates significant performance improvements over SOTA methods on both the SFDDD and AUCDD datasets. We observe a 6.5 percentage point increase in accuracy compared to DD-RCNN [53] on SFDDD, and a 0.9 percentage point improvement over C-SLSTM [57] on AUCDD. Additionally, our framework shows lower NLL loss compared to C-SLSTM, indicating more confident and accurate probabilistic predictions.

The split-by-image strategy, compared to the practical split-by-driver method, reveals a strong correlation between training and testing data. Against other SOTA models like LWANet [51] on SFDDD, our framework not only shows improved accuracy of 0.26 percentage points, but also demonstrates a notable reduction in NLL loss, proving the robustness and reliability of our model.

Furthermore, to validate the effectiveness of our MTL strategy, we conducted an ablation study comparing the MTL-enhanced ViT with a standard ViT trained solely for the action recognition task. The results, presented in Tab. III, show that our MTL framework achieves higher accuracy and lower NLL loss on both the SFDDD and AUCDD datasets. Specifically, we report accuracy improvements of 2.2 and 2.7 percentage points over the standard ViT on the SFDDD and AUCDD datasets, respectively. This highlights the benefit of incorporating emotional state recognition into the driver action recognition task, resulting in enhanced overall performance.

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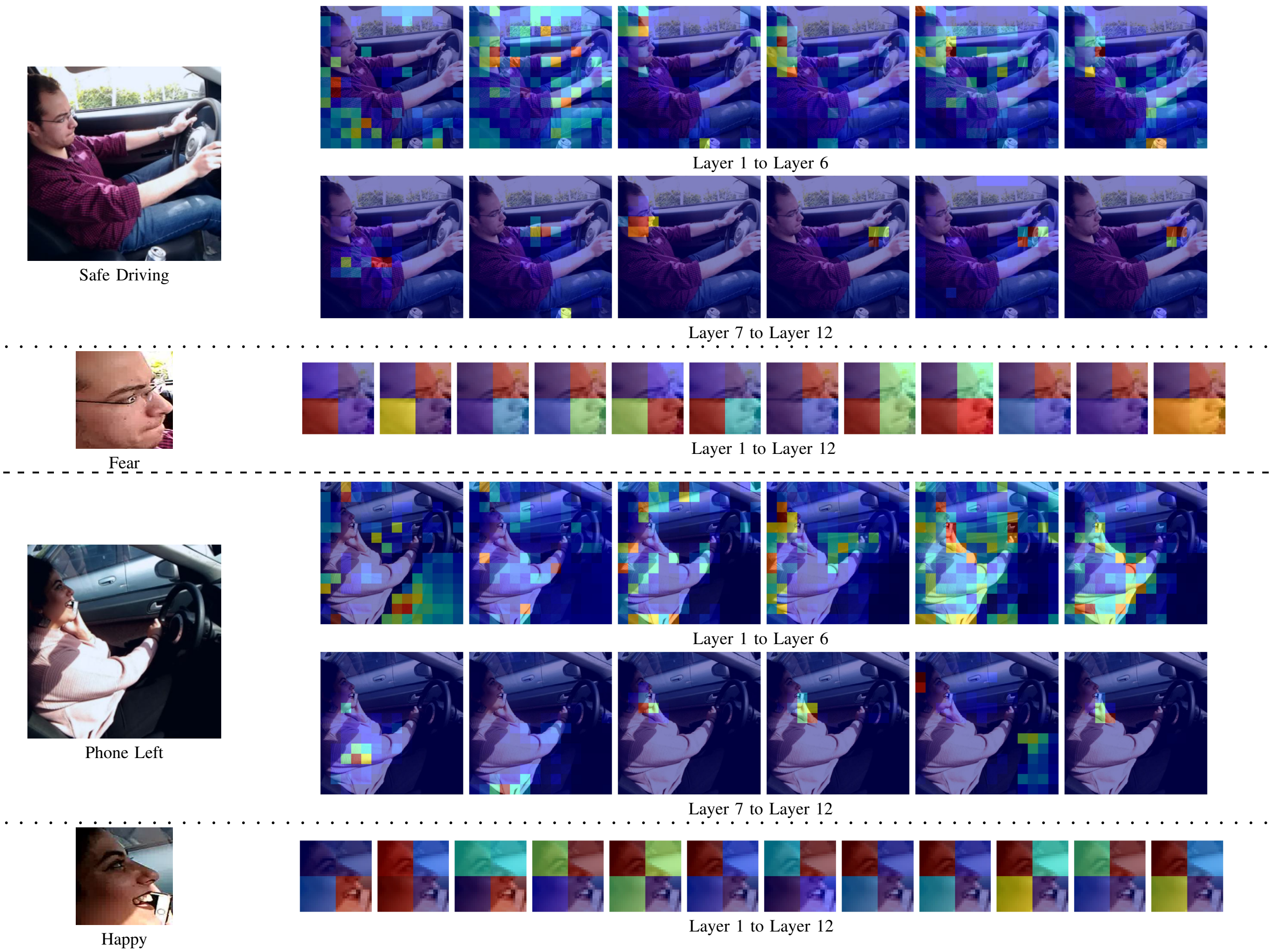


Fig. 2. This figure represents the attention maps generated by the ActionFormer during inference on the AUCDD dataset. The attention maps reveal the interplay between the class token and other visual tokens across the $L = 12$ Transformer layers. Colors in the maps correspond to the level of attention, with **red** denoting **high** attention and **blue** indicating **low** attention. As seen, the class token's focus sharpens onto precise local cues as the network delves deeper, rather than uniformly attending to the entire image. The model's ability to pinpoint vital regions within the input images, such as the steering wheel area in the first scenario and the phone region in the second, is demonstrated.

E. Results on Temporal Localization

Our proposed framework achieved an overlap score of 0.5921 on the SynDD2 test set, securing the 5th position on the public leaderboard of the 7th AI City Challenge Track 3 from among more than 200 participating teams. This performance underscores the strength of our proposed Election algorithm. To evaluate the contribution of each individual step in the algorithm, we conducted an ablation study in which we selectively removed each step one at a time, while keeping the others intact.

In scenarios where the aggregation stage was not implemented, we resorted to a baseline method that directly averaged the probability scores across the three camera views. If the filtering stage was excluded, we applied a universal threshold across all action categories to produce action candidates. Without the merging stage, the procedure advanced directly to the selection stage.

Tab. IV summarizes the results of the ablation study, presenting the scores obtained after submitting the inference results to the evaluation server. As evidenced in Tab. IV, integrating all four proposed steps resulted in the highest overlap score.

V. Visualization and Interpretability

A. Visualization of Attention Maps

To demonstrate the interpretability of our model, we generated visual attention maps during the inference process on the AUCDD dataset. These attention maps, shown in Fig. 2, illustrate the interaction between the class token and the visual tokens across different layers of the transformer encoder [58]. The attention scores are used to construct these maps. To enhance visual understanding, we reshape the 1D sequence of attention scores to match their original spatial positions in the in-cabin or face images. It is important to note that for each layer, we randomly selected one of the multi-head attentions to present the attention map.

Upon observing Fig. 2, it becomes evident that as the network goes deeper, the distraction token aggregates more

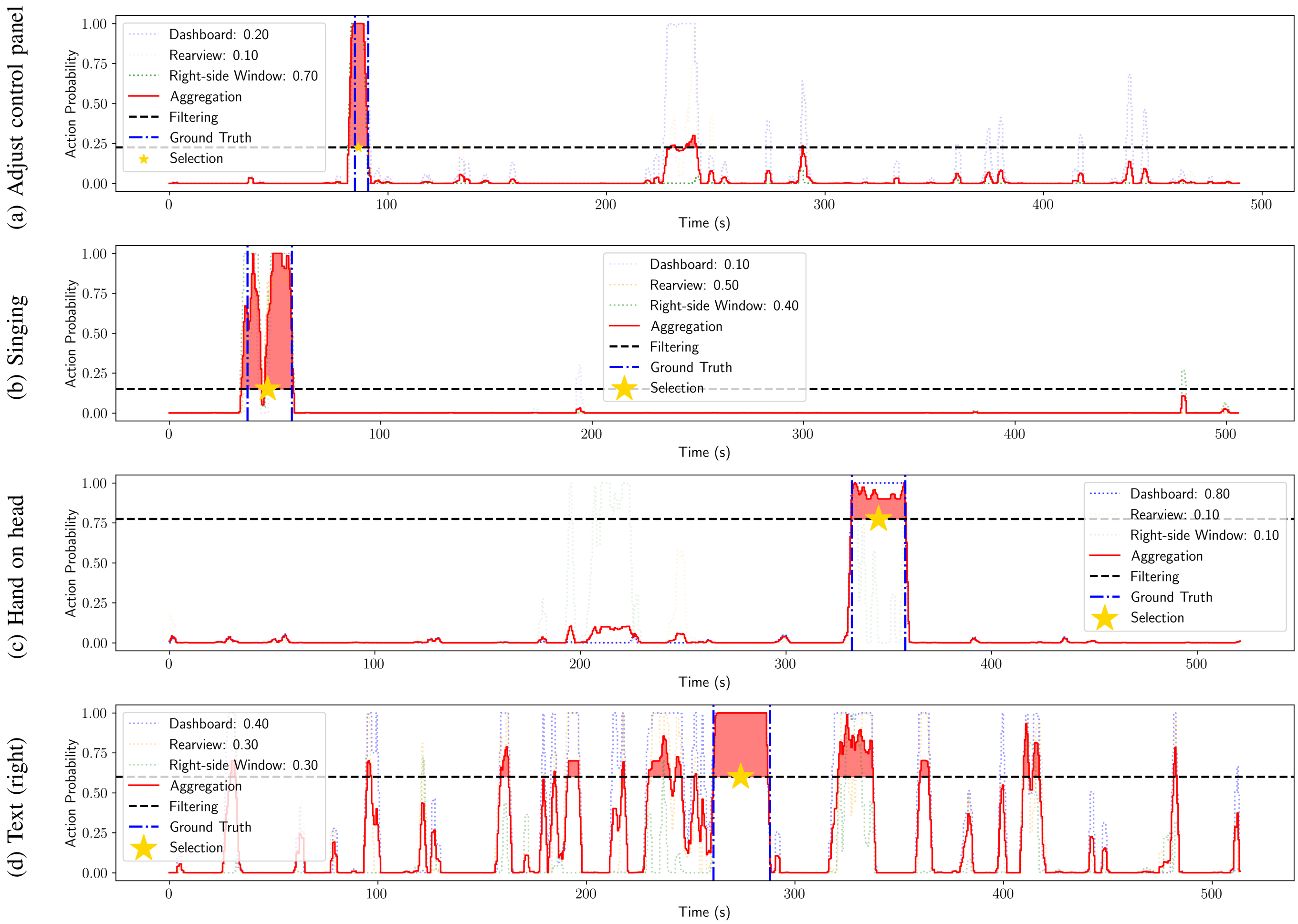


Fig. 3. This diagram demonstrates how the proposed Election algorithm incorporates initial outcomes from the action recognition module. The graph displays probability scores from the action recognition module for four action categories: (a) adjust control panel, (b) singing/dancing, (c) hand on head, and (d) text (right). The faint dotted lines represent the outputs from the recognition module, with their transparency indicating their respective contribution to identifying the action. The red line represents the probability score after the aggregation phase of the Election algorithm. The dashed black lines represent probability thresholds, while the red regions between the black dashed line and the red line indicate the potential actions. The gold star denotes the result of the selection phase.

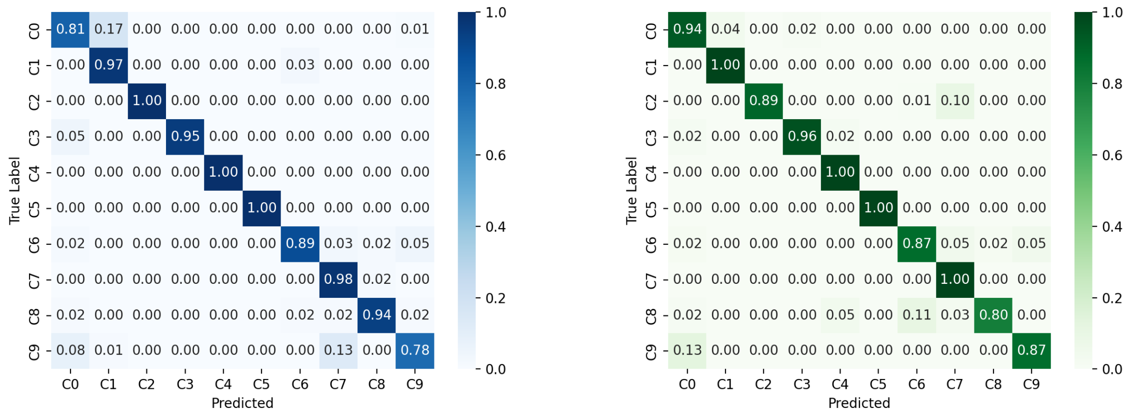
precise local cues rather than considering the entire driver or face image signals. Initially, the entire in-cabin scene contributes to interference cues, but our framework progressively focuses on important areas of the input images. For instance, in the first scenario of *safe driving*, the model successfully emphasizes the steering wheel region of the driver's image, which is the most informative part of the image. In the second scenario of *phone left*, the model effectively targets the phone region. As for the face images, the model directs attention to the eye region, which is a crucial element in recognizing drivers' facial expressions.

B. Illustrating the Election Algorithm

In Fig. 3, we illustrate how the proposed Election algorithm consolidates preliminary outcomes from the driver action recognition module within our model. The sequence of figures, arranged from top to bottom, displays the probability score indicators from the action recognition module. These are drawn from a random selection of videos from the validation set, representing four distinct action categories: *adjust control panel*, *singing or dancing with music*, *hand on head*, and *text (right)*.

In these figures, the outputs from the action recognition module are represented by the transparent dotted lines, their level of transparency correlating with their respective weights in recognizing the particular action. The solid red line indicates the probability score post the aggregation phase of the Election algorithm, while black dashed lines indicate probability thresholds. The red areas between the dashed line and the solid red line represent the potential action candidates, with ground truth start and end times of the action illustrated by blue dash-dot lines. The gold star in the figure signifies the final outcome of the selection phase.

The visual representation of the algorithm reveals the influence of various camera perspectives on the recognition of distinct behaviors. For instance, a comparison between the first and third rows illustrates that the right-side window view significantly aids in the recognition of the *adjust control panel* action (first row), but could create substantial noise when identifying the *hand on head* action (third row). This noise is subsequently rectified by the algorithm's aggregation and



(a) without MTL (b) with MTL

Fig. 4. Confusion Matrices demonstrating the framework's performance with and without the integration of MTL with Emotion Recognition.

filtering stages. Moreover, the second row, representing the *singing or dancing with music* action, exemplifies how a pause can impact recognition, a situation tackled by the algorithm's merging stage. Finally, the fourth row demonstrating the *Text (Right)* action emphasizes how the selection stage thoroughly assesses all merged candidates along with their scores to deliver a holistic final prediction.

This visualization illustrates how our proposed algorithm effectively improves the accuracy of temporal action localization in multi-view videos by leveraging the complementary information from different camera views and selecting the most reliable action candidates. The integration of convolution kernels tailored for specific action categories and the merging of overlapping candidates overcomes the limitations of individual camera views and effectively captures the temporal characteristics of the actions, resulting in improved temporal action localization performance.

C. Insights from Emotion Recognition

The confusion matrices for the AUCDD dataset, both with and without the MTL module, are presented in Fig. 4. These matrices yield several important observations:

(1) The presence of MTL significantly enhances the framework's ability to detect *safe driving* and *talk to passenger*, with improvements in the accuracy of 13 and 9 percentage points respectively. This enhancement is due to the strong correlation between certain emotional states and these two types of driving behavior. For instance, drivers often exhibit a neutral emotion during safe driving and a happy emotion when conversing with passengers. Without MTL, *safe driving* is often misinterpreted as *phone call (R)*, a mistake which can be mitigated through the use of driver emotion information, given that phone conversations can trigger a variety of emotional states, not just neutral ones. Additionally, in the absence of MTL, *talk to passenger* is misclassified as *hair or makeup* in 13% of instances, a mistake eliminated thanks to the inclusion of emotion information.

(2) On the downside, MTL introduces a decrement in performance when detecting *phone call (L)* and *reaching behind*. Specifically, *phone call (L)* is occasionally misinterpreted as *hair or makeup*, while *reaching behind* is sometimes wrongly classified as *drinking*. In both these scenarios, emotion information could potentially mislead the detection of driver distractions: a wide range of emotions can be associated with phone use, rendering emotional data unhelpful for detecting this behavior; as for *reaching behind*, discerning the driver's emotion can be challenging, leading to potential inaccuracies in the emotion data.

(3) Notably, our framework demonstrates superior detection accuracy for *phone call (R)* over *phone call (L)*, a discrepancy likely attributable to the inherent bias in the dataset. Given that the camera is recording the driver from the right-hand side, detecting phone usage on the right is considerably easier than on the left, as the phone on the right is entirely visible.

VI. Conclusions and Future Work

In this study, we have introduced a comprehensive framework within the Driver Digital Twin (DDT) concept. This framework effectively uses both the physical behavior and emotional state of the driver as indicators of distraction. To achieve this, we have used a transformer-based driver action recognition model called the ActionFormer. To improve the accuracy of the ActionFormer's initial findings, we have incorporated a new Election algorithm specifically designed to refine its outputs. Additionally, the combination of multi-task learning (MTL) and emotion recognition mechanisms has significantly improved the overall distraction detection capability. The effectiveness of our proposed framework has been thoroughly validated against three distraction detection benchmarks, including SFDDD, AUCDD, and SynDD2. Our results clearly demonstrate its superior performance compared to existing distraction detection frameworks.

While our research represents a significant advancement in the DDT paradigm, we acknowledge the importance of further development and improvement. In this regard, our future efforts will focus on several key areas of improvement:

a) Optimization of the use of emotion information: We will explore techniques to better separate different emotional states within the MTL model. By distinguishing between emotions that genuinely indicate distraction and those that do not, we aim to reduce the risk of emotional information leading to false positives or misclassification. To ensure a balanced dataset and minimize the risk of emotional bias, we will consider data augmentation techniques and strategies that provide a more representative distribution of emotional states during the distraction detection process. We will conduct user studies and gather feedback from real-world drivers to gain insights into the impact of emotional states during driving and their relevance to distraction. This user-centric approach will guide our optimization efforts.

b) Integration into vehicular interfaces: We aim to integrate distraction prediction models into vehicular interfaces to improve driving safety and experience [64]–[67]. User-friendly interfaces provide real-time feedback based on distraction predictions. For example, if signs of distraction are detected, the interface can issue alerts or adjust the vehicle's settings. Personalized recommendations can be tailored to the driver's behavior and emotional state. A robust communication framework ensures quick response times and minimal latency in delivering alerts and interventions.

c) Scalability and transferability: We aim to develop models that are widely applicable across various vehicles, considering computational efficiency and hardware limitations [68], [69]. The proposed model needs to be optimized for efficiency and hardware variations. A standardized API or interface should be developed for easy integration into different vehicle makes and models. Collaboration with automakers and adherence to industry standards are crucial. Rigorous testing and validation procedures ensure reliable performance in different driving conditions, vehicles, and demographics.

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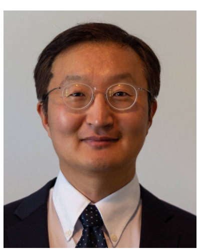

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